MTH 202 Chapter 5: Topics In Vector Calculus

5.2 Line Integrals Worksheet

1. Let C be the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t)=\langle \sin(t), 2t, \cos(t)\rangle$ for $0 \le t \le \dfrac{\pi}{2}$.

a) Evaluate the integral $\int_C x^2 z\, ds$.

$$\int_C x^2 z\, ds = \int_0^{\pi/2} \sin^2(t)\cos(t)\sqrt{\cos^2(t)+4+\sin^2(t)}\, dt = \int_0^{\pi/2} \sin^2(t)\cos(t)\sqrt{\cos^2(t)+4+\sin^2(t)}\, dt$$

$$= \int_0^{\pi/2} \sin^2(t)\cos(t)\sqrt{5}\, dt = \frac{\sqrt{5}}{3}\sin^3(t)\Big|_0^{\pi/2} = \frac{\sqrt{5}}{3}(1-0) = \frac{\sqrt{5}}{3}$$

b) Evaluate the integral $\int_C x^2 z\, dx$.

$$\int_C x^2 z\, dx = \int_0^{\pi/2} \sin^2(t)\cos(t)\cos(t)\, dt = \int_0^{\pi/2} \sin^2(t)\cos^2(t)\, dt = \int_0^{\pi/2} \frac{1}{2}\left[1-\cos(2t)\right]\frac{1}{2}\left[1+\cos(2t)\right] dt$$

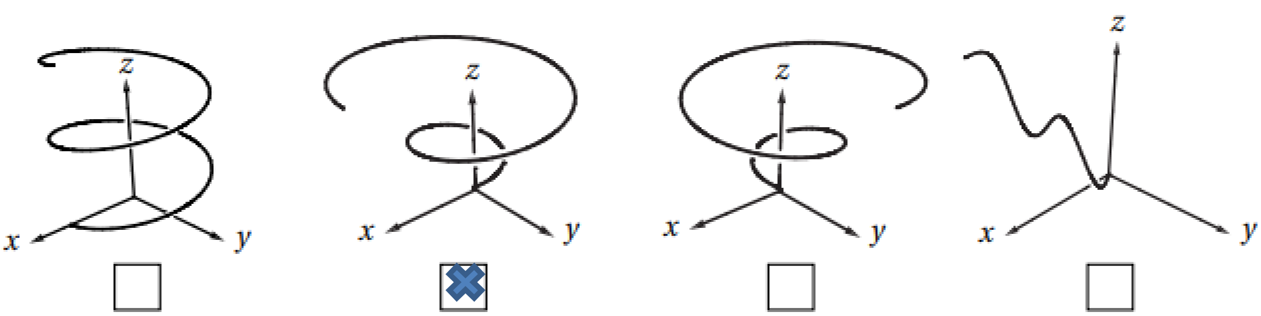
$$= \frac{1}{4}\int_0^{\pi/2}\left[1-\cos^2(2t)\right] dt = \frac{1}{4}\int_0^{\pi/2}\left[1-\frac{1}{2}\left[1+\cos(4t)\right]\right] dt = \frac{1}{4}\int_0^{\pi/2}\left[\frac{1}{2}-\frac{1}{2}\cos(4t)\right] dt$$

$$= \frac{1}{4}\left(\frac{1}{2}t-\frac{1}{8}\sin(4t)\right)\Bigg|_0^{\pi/2} = \frac{1}{4}\left[\left(\frac{\pi}{4}-0\right)-(0-0)\right] = \frac{\pi}{16}$$

c) Evaluate the integral $\int_C x^2 z\, dy$.

$$\int_C x^2 z\, dy = \int_0^{\pi/2} \sin^2(t)\cos(t)\, 2\, dt = \frac{2}{3}\sin^3(t)\Big|_0^{\pi/2} = \frac{2}{3}(1-0) = \frac{2}{3}$$

2. Mark the picture of the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t)=\langle t\cos(t), t\sin(t), t\rangle$ for $0 \le t \le 4\pi$.



3. Let C be the curve parameterized by $\mathbf{r}(t)=\langle\cos(t),\sin(t),2t\rangle$ for $0\le t\le 2\pi$. Compute $\int_C z\,ds$.

We see that

$$\int_C z\,ds=\int_0^{2\pi}(2t)\sqrt{\sin^2(t)+\cos^2(t)+4}\,dt=\int_0^{2\pi}(2t)\sqrt{5}\,dt=t^2\sqrt{5}\Big|_0^{2\pi}=4\pi^2\sqrt{5}$$

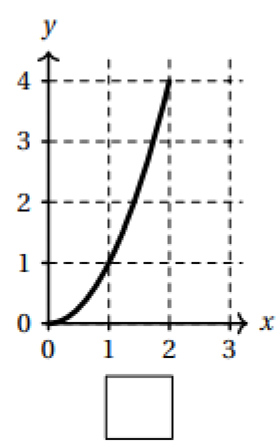
4. Find the mass of a thin wire in the shape of the curve $\mathbf{r}(t)=\langle\sin(t),2\sin(t),\sqrt{5}\cos(t)\rangle$ where $0\le t\le\pi$, if the wire has density function $\rho(x,y,z)=x$.

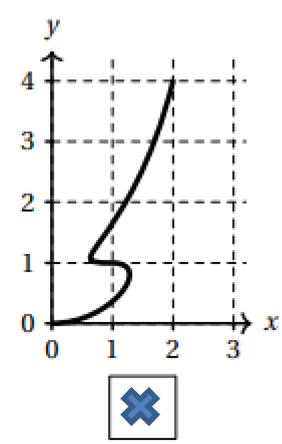
$$mass=\int_C\rho(x,y,z)\,ds=\int_0^{\pi}\sin(t)\sqrt{\cos^2(t)+4\cos^2(t)+5\sin^2(t)}\,dt$$

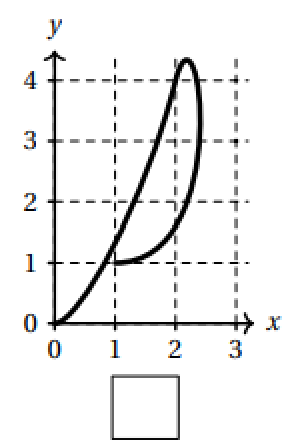
$$=\int_0^{\pi}\sin(t)\sqrt{5}\,dt=-\sqrt{5}\cos(t)\Big|_0^{\pi}=-\sqrt{5}(-1-1)=2\sqrt{5}$$

5. Suppose that $\mathbf{r}:[0,2]\to\mathbb{R}^2$ is a parametric curve in the plane and that $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ have the values given in the table at left. Check the box below the picture that could be a plot of $\mathbf{r}(t)$.

t	$\mathbf{r}(t)$	$\mathbf{r}'(t)$
0	$(0,0)$	$\mathbf{i}$
1	$(1,1)$	$-\mathbf{i}$
2	$(2,4)$	$\mathbf{i}+4\mathbf{j}$

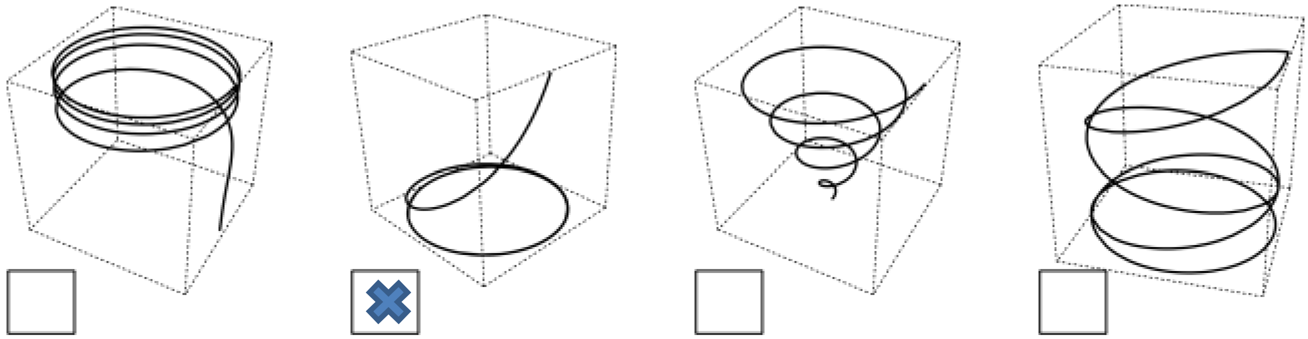






6. Let C be the oriented curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), e^t \rangle$ where $0 \le t \le 8\pi$.

a) Check the box next to the picture which best matches C.



b) Calculate the line integral $\int_C z\,dz$

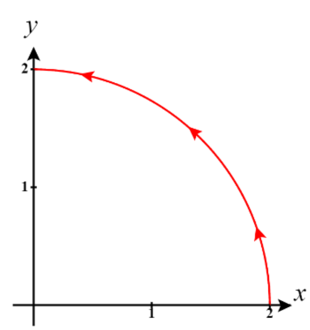
$$\int_C z\,dz = \int_0^{8\pi} e^t e^t\,dt = \int_0^{8\pi} e^{2t}\,dt = \frac{1}{2}e^{2t}\Big|_0^{8\pi} = \frac{1}{2}\left(e^{16\pi} - 1\right)$$

7. Suppose a wire can be modeled by the segment of the circle $x^2 + y^2 = 4$ from $(2,0)$ to $(0,2)$, when moving counterclockwise. If the linear density of this wire is measured by $f(x,y) = xy$ for any point on the wire, then set up an integral which represents the mass of this wire.

We know that the mass of the wire is given by $M = \int_C xy\,ds$ where C can be parameterized at

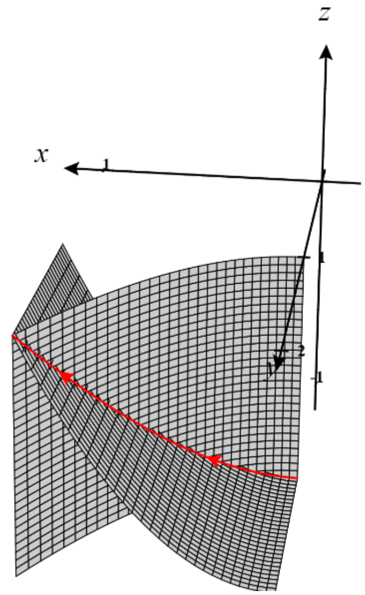
$\mathbf{r}(t) = \langle 2\cos(t), 2\sin(t) \rangle$ where $0 \le t \le \frac{\pi}{2}$. Therefore, we calculate that

$$\begin{aligned} M &= \int_C xy\,ds = \int_0^{\pi/2} 4\cos(t)\sin(t)\sqrt{4\cos^2 + 4\sin^2(t)}\,dt \\ &= 8\int_0^{\pi/2} \cos(t)\sin(t)\,dt = 4\sin^2(t)\Big|_0^{\pi/2} \\ &= 4(1) - 4(0) \\ &= 4 \end{aligned}$$



8. Let $f(x,y,z) = xy - xz$, and C be a curve which is the intersection of the surfaces $y = x^2 + 1$ and $z = x^2 - 1$ from $(0,1,-1)$ to $(1,2,0)$. Compute the line integral $\int_C f(x,y,z)\,ds$.

We can see that the intersection curve C can be easily parameterized as $\mathbf{r}(t) = \langle t, t^2+1, t^2-1 \rangle$ where $0 \le t \le 1$.

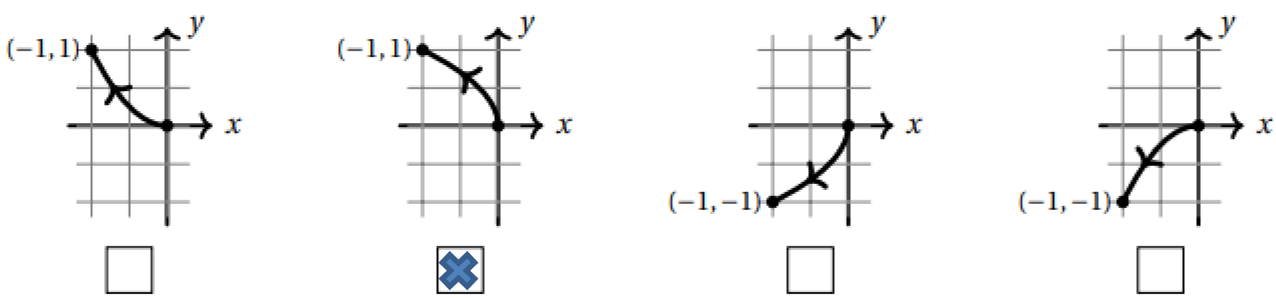


Therefore, we compute that

$$\begin{aligned}
\int_C f(x,y,z)\,ds &= \int_0^1 \left(t\left(t^2+1\right) - t\left(t^2-1\right)\right)\sqrt{1^2 + (2t)^2 + (2t)^2}\,dt \\
&= \int_0^1 2t\sqrt{8t^2+1}\,dt = \frac{1}{12}\left(1+8t^2\right)^{3/2}\Big|_0^1 \\
&= \frac{1}{12}(9)^{3/2} - \frac{1}{12}(1)^{3/2} = \frac{1}{12}(27-1) \\
&= \frac{13}{6}
\end{aligned}$$

9. Let C be the curve in $\mathbb{R}^2$ parameterized by $\mathbf{r}(t) = \langle -t^2, t \rangle$ for $0 \le t \le 1$.

a) Mark the picture of C from among the choices below.

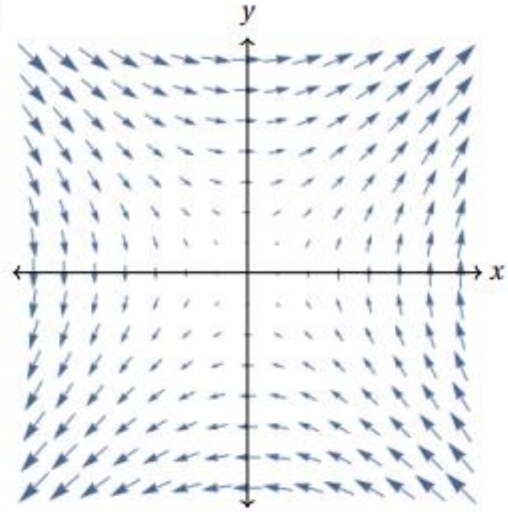


b) For the vector field $\mathbf{F} = \langle y, -x \rangle$ compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

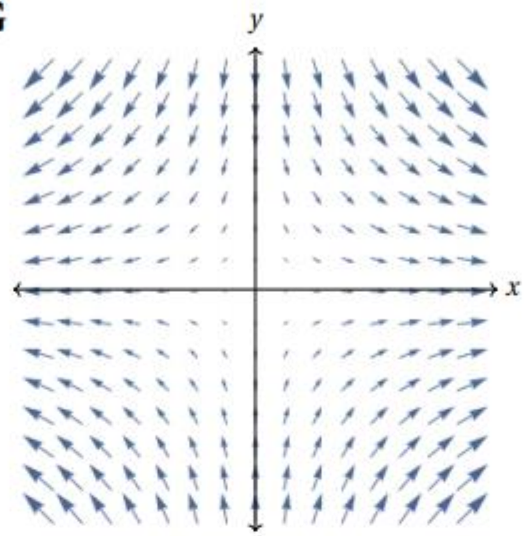
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \langle t, t^2 \rangle \cdot \langle -2t, 1 \rangle\,dt = \int_0^1 \left(-2t^2 + t^2\right)dt = \int_0^1 \left(-t^2\right)dt = -\frac{1}{3}t^3\Big|_0^1 = -\frac{1}{3}$$

10. Three continuous vector fields **F**,**G**, **H** on $\mathbb{R}^2$ are plotted below in the region with $-1 < x < 1$ and $-1 < y < 1$. Circle the best response for each of the following questions.

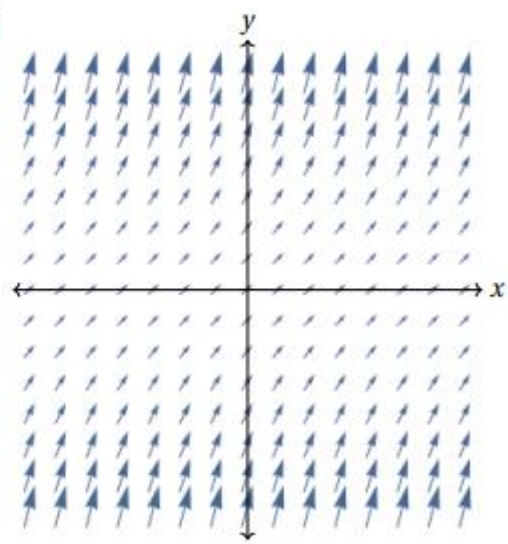
F



G



H



a) Which vector field represents $\vec{\nabla} f$, where $f(x, y) = xy$.

F	**G**	**H**

We see that $\vec{\nabla} f = \langle y, x \rangle$ which is represented by **F**.

b) A particle moves along a straight line from $(-1,-1)$ to $(1,1)$. If the vector fields represent force fields, then:

The work done by **G** on the particle is…

positive	negative	zero

At every point on the line the direction of the vector field is orthogonal to the direction vector of the line and hence the work done is 0.

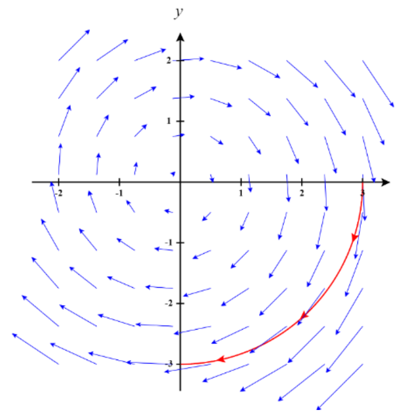
The work done by **H** on the particle is…

positive	negative	zero

At every point on the line the direction of the vector field makes an angle of less than $\frac{\pi}{2}$ with the direction vector of the line. Therefore, the work done is positive.

11. Find $\int_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ and C is the path along the circle of radius 3 starting at the point $(3,0)$ and ending at the point $(0,-3)$ while moving clockwise.

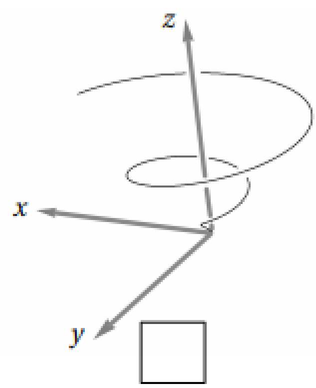
We begin by parameterizing the path C. One possible parameterization would be $\mathbf{r}(t) = \langle 3\cos(t), -3\sin(t)\rangle$ where $0 \le t \le \frac{\pi}{2}$.

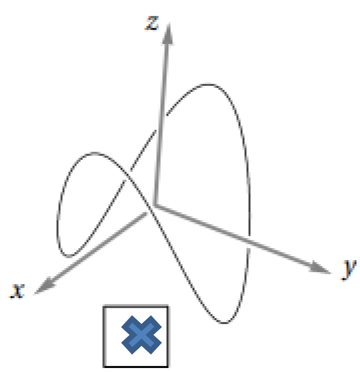


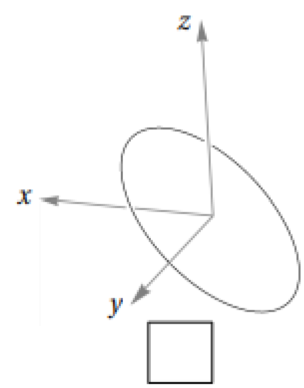
$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{\pi/2} \langle -3\sin(t), -3\cos(t)\rangle \cdot \langle -3\sin(t), -3\cos(t)\rangle\, dt$$
$$= \int_0^{\pi/2} \left[9\sin^2(t) + 9\cos^2(t)\right] dt = \int_0^{\pi/2} 9\,dt$$
$$= \frac{9\pi}{2}$$

12. Consider the curve C parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t)\rangle$ where $0 \le t \le 2\pi$.

a) Check the box next to the correct sketch of C.







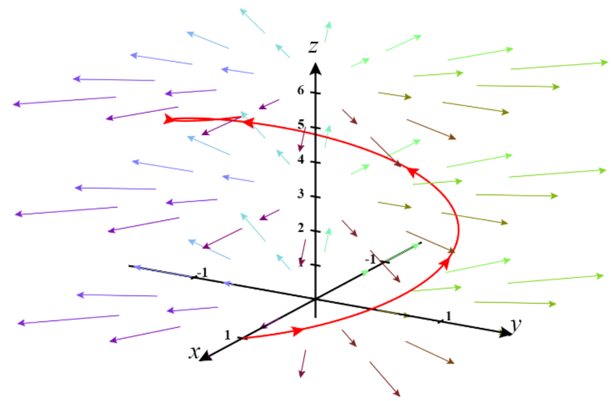
b) Find the work done if a particle travels along path $\mathbf{r}(t)$ under the force given by the vector field $\mathbf{F}(x,y,z) = \langle -2y, 2x, 0\rangle$.

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} \langle -2\sin(t), 2\cos(t), 0\rangle \cdot \langle -\sin(t), \cos(t), -2\sin(2t)\rangle\, dt$$
$$= \int_0^{2\pi} 2\left[\sin^2(t) + \cos^2(t)\right] dt$$
$$= 4\pi$$

13. Let C be the helix parameterized by $\mathbf{r}(t)=\langle\cos(t),\sin(t),t\rangle$ where $0\le t\le 2\pi$.

a) Let $\mathbf{F}(x,y,z)$ be the vector field defined by $\mathbf{F}(x,y,z)=\langle x,y,0\rangle$. Determine $\int_C \mathbf{F}\cdot d\mathbf{r}$ when C is oriented in the direction from $(1,0,0)$ to $(1,0,2\pi)$.

$$
\begin{aligned}
\int_C \mathbf{F}\cdot d\mathbf{r} &= \int_0^{2\pi}\langle\cos(t),\sin(t),0\rangle\cdot\langle-\sin(t),\cos(t),1\rangle\,dt\\
&= \int_0^{2\pi}\left[-\sin(t)\cos(t)+\sin(t)\cos(t)+0\right]dt\\
&= \int_0^{2\pi} 0\,dt\\
&= 0
\end{aligned}
$$



b) Now let $\mathbf{F}(x,y,z)=\left\langle ye^{x^2+y^2},\ln\left(x^2+y^2\right)-x,1\right\rangle$. Determine $\int_C \mathbf{F}\cdot d\mathbf{r}$ when C is given the opposite orientation.

$$
\begin{aligned}
\int_{-C} \mathbf{F}\cdot d\mathbf{r} &= -\int_C \mathbf{F}\cdot d\mathbf{r}\\
&= -\int_0^{2\pi}\langle\sin(t)e^1,\ln(1)-\cos(t),1\rangle\cdot\langle-\sin(t),\cos(t),1\rangle\,dt\\
&= -\int_0^{2\pi}\left[-\sin^2(t)e-\cos^2(t)+1\right]dt\\
&= \int_0^{2\pi}\left[\sin^2(t)e+\cos^2(t)-1\right]dt\\
&= \int_0^{2\pi}\left[\frac{e}{2}\left[1-\cos(2t)\right]+\frac{1}{2}\left[1+\cos(2t)\right]-1\right]dt\\
&= \left[\frac{e}{2}\left[t-\frac{1}{2}\sin(2t)\right]+\frac{1}{2}\left[t+\frac{1}{2}\sin(2t)\right]-t\right]_0^{2\pi}\\
&= \left[\frac{et}{2}-\frac{e}{4}\sin(2t)+\frac{t}{2}+\frac{1}{4}\sin(2t)-t\right]_0^{2\pi}\\
&= \left[\pi e-0+\pi+0-2\pi\right]-\left[0-0+0+0+0\right]\\
&= \pi e-\pi\\
&= \pi(e-1)
\end{aligned}
$$